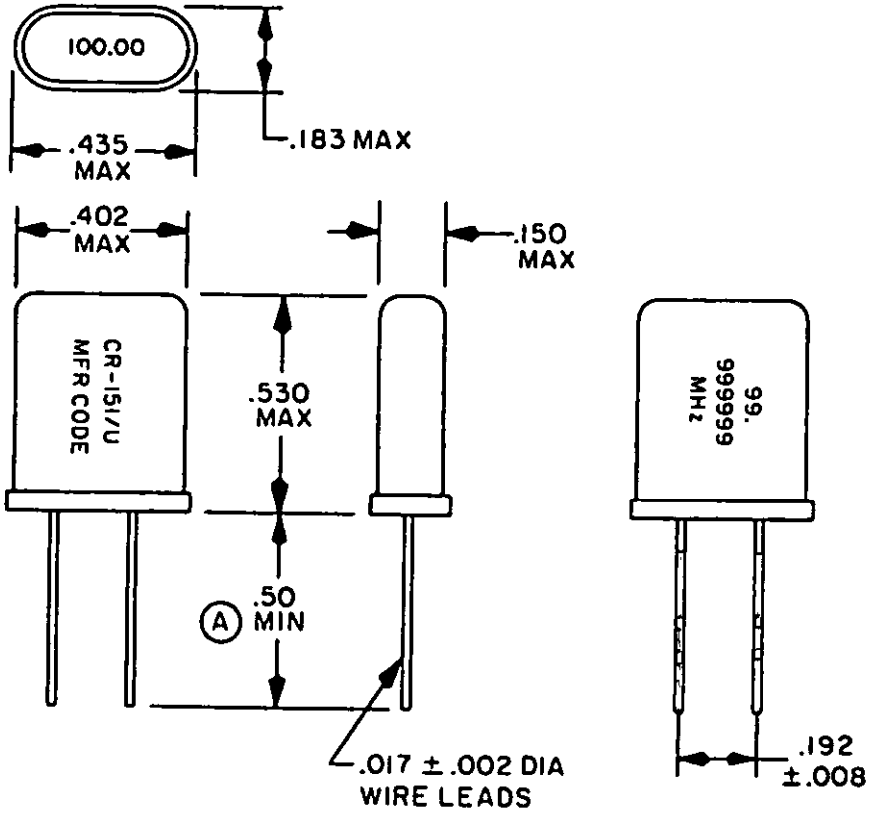
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-151/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 50 to 100 MHz; fifth mechanical overtone; noncontrolled; series resonance.



INCHES	MM
.002	.05
.008	.20
.017	.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.50	12.7
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. The type designator, frequency, and manufacturer's code shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.

FIGURE 1. Crystal unit - CR-151/U.

Ⓐ denotes changes

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FSC 5955

Ⓐ TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-18/U or HC-43/U (MIL-H-10056/9)
Dimensions, marking, and configuration	See figure 1.
Frequency range	50 to 100 MHz, inclusive
Capacitance, shunt	6.0 pF +0.5, -1.0 pF
Mode of oscillation	Fifth mechanical overtone
Government reference standard test set	TS-683/TSM
Rated drive level	2.0 mW, maximum
Calibration values:	
Resistance	40 ohms
Resistor voltage drop	0.28 volt
Operating temperature ranges:	
Room temperature	25°C to 30°C, inclusive
Primary	-30°C to 25°C and 30°C to 70°C, inclusive
Secondary	-40°C to +30°C and 70°C to 90°C, inclusive
Frequency tolerances:	
Room temperature	±20 ppm
Primary operating temperature range	±35 ppm
Secondary operating temperature range	±50 ppm
Equivalent resistance	50 ohms, maximum
Shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Vibration: 1/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Thermal shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	±5 ppm

1/ Method 204 of MIL-STD-202, test condition A.

NOTE: These crystal units are for use in the AN/ARC-101 equipment.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 80
DLA - ES

User activities:
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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